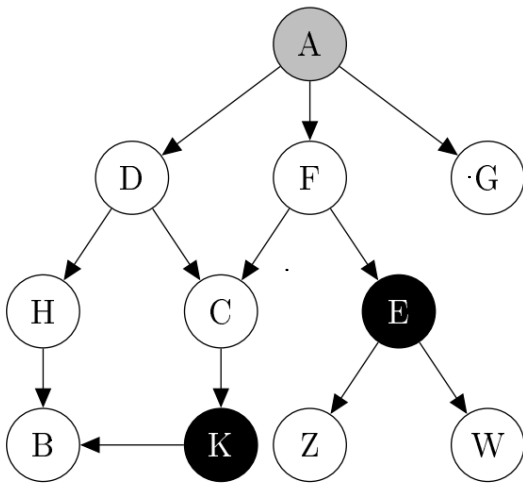


2.a)



Solución: $A \rightarrow F \rightarrow E$

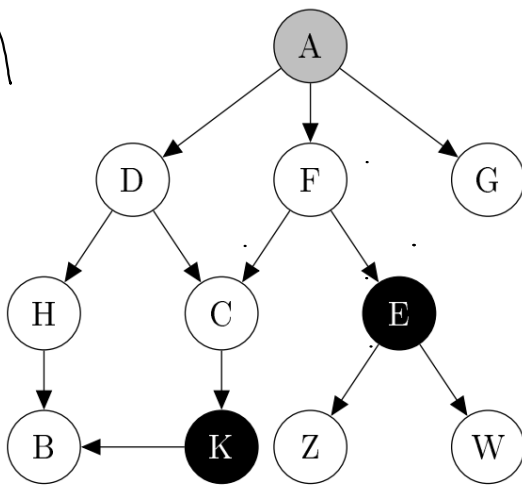
Abiertos

~~$(-, A)$~~
 ~~(A, D)~~
 ~~(A, F)~~
 (A, G)
 (D, H)
 (D, C)
 (F, C)
 $(F, E) \rightarrow$ estado final

Cerrados

~~$(-, A)$~~
 ~~(A, D)~~
 ~~(A, F)~~

b)



Abiertos

~~$(-, A)$~~
 (A, D)

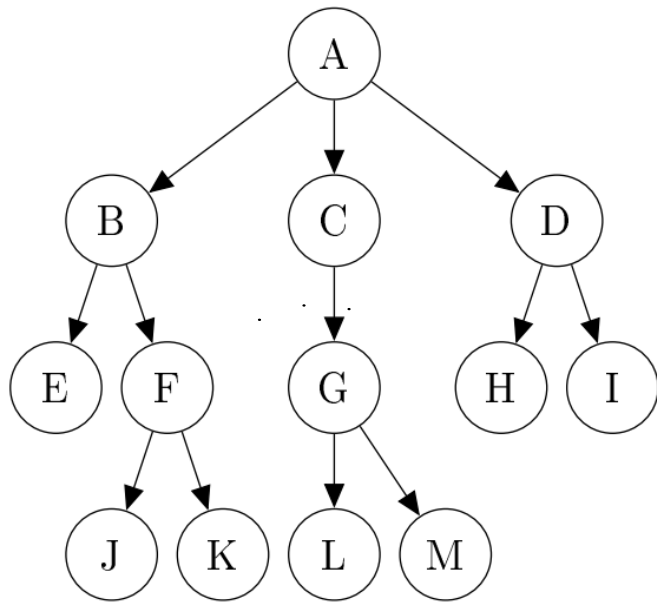
Cerrados

~~$(-, A)$~~

~~$(-, A)$~~ , (A, D) , (D, H) , ~~(H, B)~~ , (D, C) , $(C, K) \rightarrow$ estado final

Solución: A, D, C, K

3.

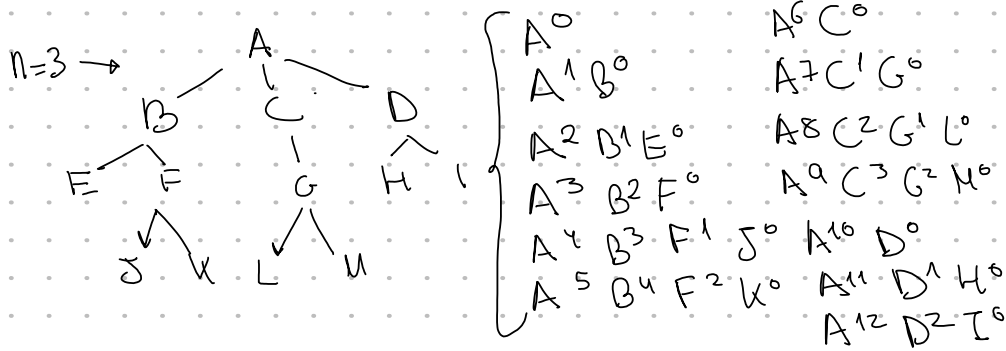
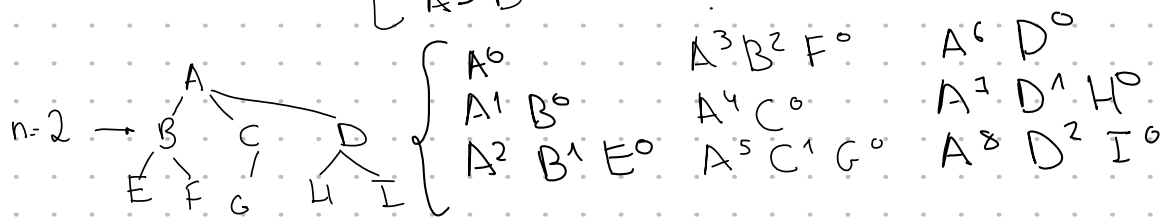
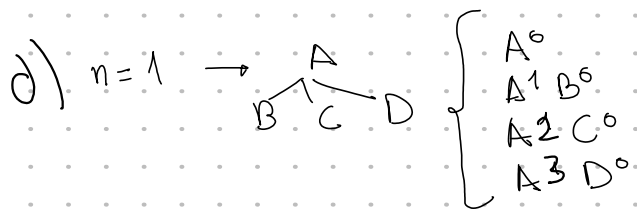


a) A, B, C, D, E, F, G, H, I, J, K, L, M

b) A, B, E, F, J, K, C, G, L, M, D, H, I

c) A A, C, G
 A, B A, C
 A, B, E
 A, B
 A, B, F
 A, B, F, J
 A, B, F
 A, B, F, K
 A, B, F
 A, B
 A
 A, C
 A, C, G
 A, C, G, L
 A, C, G
 A, C, G, M

A, C, G
 A, C
 A
 A, D
 A, D, H
 A, D
 A, D, I
 A, D
 A



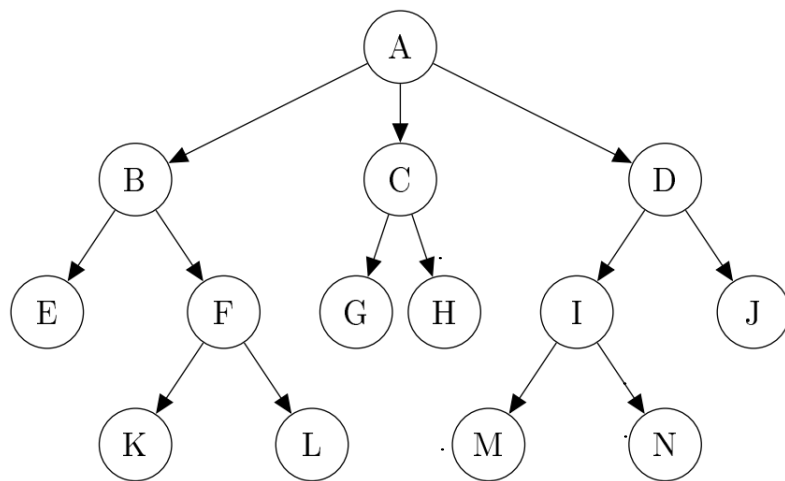
Sequencia: A, B, C, D

A, B, E, F, C, G, D, H, I

A, B, E, F, J, K, C, G, L, M, D, H, I

Sewencia: A, B, C, D, E, F, G, H, I, J, K, L, M
sin repetir

4.



a) anchura

A, B, C, D, E, F, G, H, I, J, K, L, M, N

b) retroactiva con profundidad limitada ($\delta=2$)

A	A	A	A
A, B	A, C	A, D	
A, B, E	A, C, G	A, D, I	
A, B	A, C	A, D	
A, B, F	A, C, H	A, D, J	
A, B	A, C	A, D	

Orden en el que se visitan los nodos:

A, B, E, F, C, G, H, D, I, J

c) descenso iterativo (profundidad inicial = 1)

$\delta=1 \rightarrow$ $\left\{ \begin{array}{l} A^0 \\ A^1 B^0 \\ A^2 C^0 \\ A^3 D^0 \end{array} \right.$

$\delta=2 \rightarrow$ $\left\{ \begin{array}{l} A^0 \\ A^1 B^0 \\ A^2 B^1 E^0 \\ A^3 B^2 F^0 \end{array} \right.$ $\left\{ \begin{array}{l} A^4 C^0 \\ A^5 C^1 G^0 \\ A^6 C^2 H^0 \end{array} \right.$ $\left\{ \begin{array}{l} A^7 D^0 \\ A^8 D^1 I^0 \\ A^9 D^2 J^0 \end{array} \right.$

$\delta=3 \rightarrow$ $\left\{ \begin{array}{l} A^0 \\ A^1 B^0 \\ A^2 B^1 E^0 \\ A^3 B^2 F^0 \\ A^4 B^3 F^1 K^0 \\ A^5 B^4 F^2 L^0 \end{array} \right.$ $\left\{ \begin{array}{l} A^6 C^0 \\ A^7 C^1 G^0 \\ A^8 C^2 H^0 \end{array} \right.$ $\left\{ \begin{array}{l} A^9 D^0 \\ A^{10} D^1 I^0 \\ A^{11} D^2 I^1 M^0 \\ A^{12} D^3 I^2 N^0 \\ A^{13} D^4 J^0 \end{array} \right.$

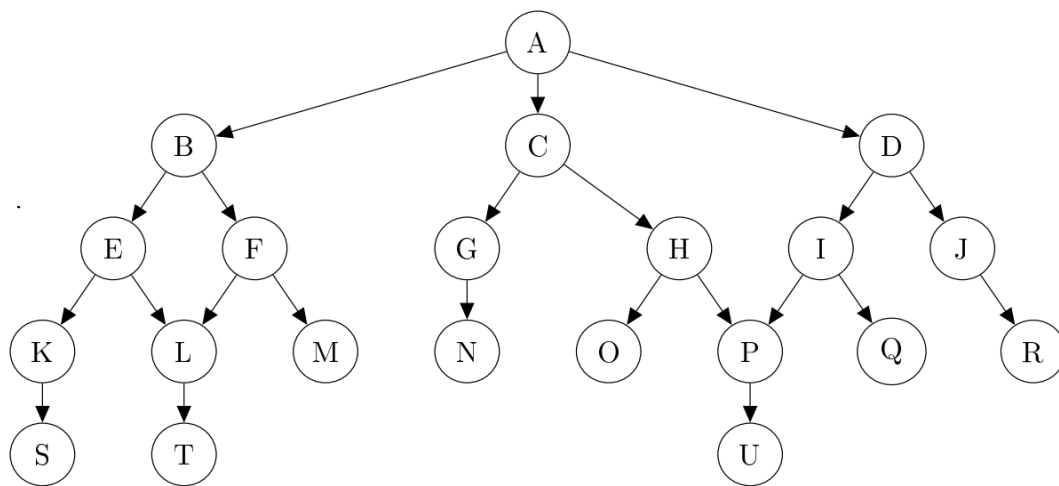
Orden de exploración:

A, B, C, D, A, B, E, F, C, G, H, D, I, J, A, B, E, F, K, L, C, G, H, D, I, M, N, J

Orden en el que se visitan los nodos:

A, B, C, D, E, F, G, H, I, J, K, L, M, N

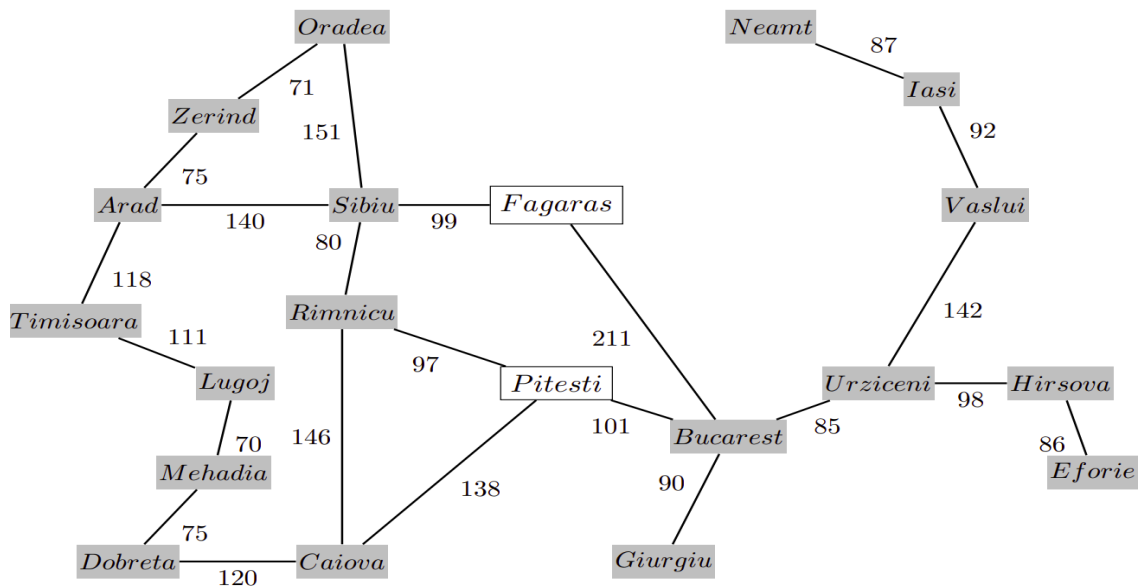
5.



Búsqueda retroactiva:

A, AB, ABE, ABEK, ABEKS, ABEK, ABE, ABEL, ABELT, ABEL, ABE, AB, ABF, ABFL, ABFLT, ABFL, ABF, ABFM, ABF, AB, A, AC, ACG, ACGN, ACG, AC, ACH, ACHO, ACH, ACHP, ACHPU, ACHP, ACH, AC, AAD, ADI, ADIP, ADIPU, ADIP, ADI, ADIQ, ADI, AD, ADS, ADSR, ADS, AD, A

10.
coste
uniforme
desde
Pitesti
hasta
Fagaras



Abiertos

~~(-, P, 0)~~

~~(P, R, 97)~~

~~(P, C, 138)~~

~~(P, B, 101)~~

~~(R, S, 177)~~

~~(R, C, 243)~~

~~(B, F, 312)~~

~~(B, G, 191)~~

~~(B, U, 186)~~

~~(C, R, 284)~~

(C, D, 258)

(S, F, 276)

(S, O, 328)

(S, A, 317)

(U, V, 328)

(U, H, 284)

(D, U, 333)

Camino:
P, R, S, F

Coste:
276

Cerrados

(-, P, 0)

(P, R, 97)

(P, B, 101)

(P, C, 138)

(R, S, 177)

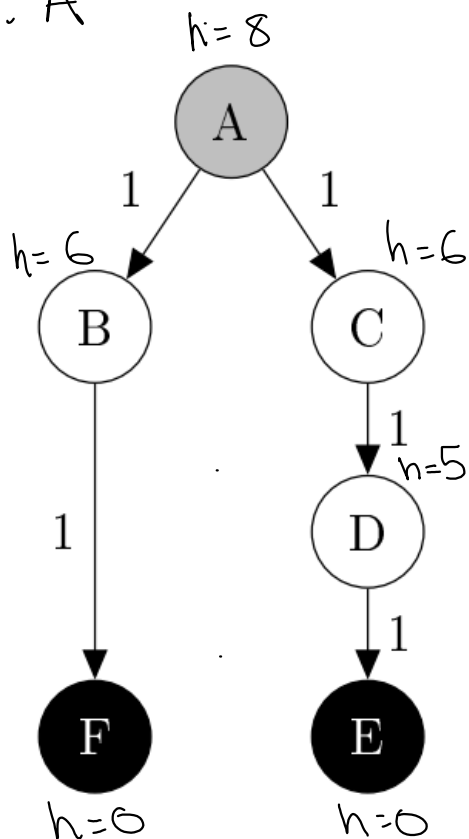
(B, U, 186)

(B, G, 191)

(C, D, 258)

(S, F, 276)

14. A*



	A	B	C	D	E	F
<i>h</i>	8	6	6	5	0	0

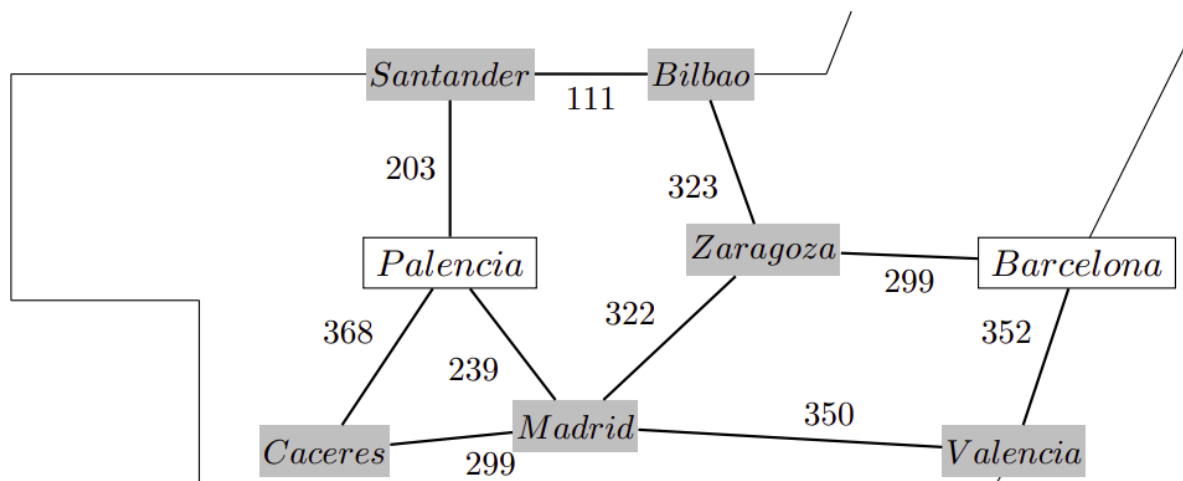
Abiertos
~~(-, A, 8)~~
~~(A, B, 7)~~
 (A, C, 7)
(B, F, 2)

Cerrados
 (-, A, 8)
 (A, B, 7)

No se continúa explorando
 ni se explora C porque
 ya se ha encontrado un
 nodo meta con el menor
 valor $f(n) = g(n) + h(n)$

- b) El orden de expansión de los nodos es el mismo orden que aparece en Cerrados, ya que un nodo entra en Cerrados precisamente cuando se ha acabado de expandir. De esta manera, los nodos que se expanden son A y B, en ese orden.
- c) Se encuentra primero el nodo meta F
- d) No, porque en todo momento sobreestima el coste real desde el nodo actual hasta el nodo final

17.



a) $h(n) = 0 \rightarrow$ coste uniforme

Abiertos

~~(-, P, 0)~~

~~(P, C, 368)~~

~~(P, M, 239)~~

~~(P, S, 203)~~

~~(S, B, 314)~~

~~(M, C, 538)~~

~~(M, V, 589)~~

~~(M, Z, 561)~~

~~(B, Z, 637)~~

~~(C, M, 667)~~

(Z, Bar, 860)

~~(V, Bar, 941)~~

Cerrados

(-, P, 0)

(P, S, 203)

(P, M, 239)

(S, B, 314)

(P, C, 368)

(M, Z, 561)

(M, V, 589)

(Z, Bar, 860)

6)

	Bilbao	Cáceres	Madrid	Palencia	Santander	Valencia	Zaragoza
Barcelona	502	850	550	580	605	303	275

Abiertos~~(-, P, 580)~~~~(P, C, 1218)~~~~(P, M, 789)~~~~(P, S, 808)~~~~(M, C, 1388)~~

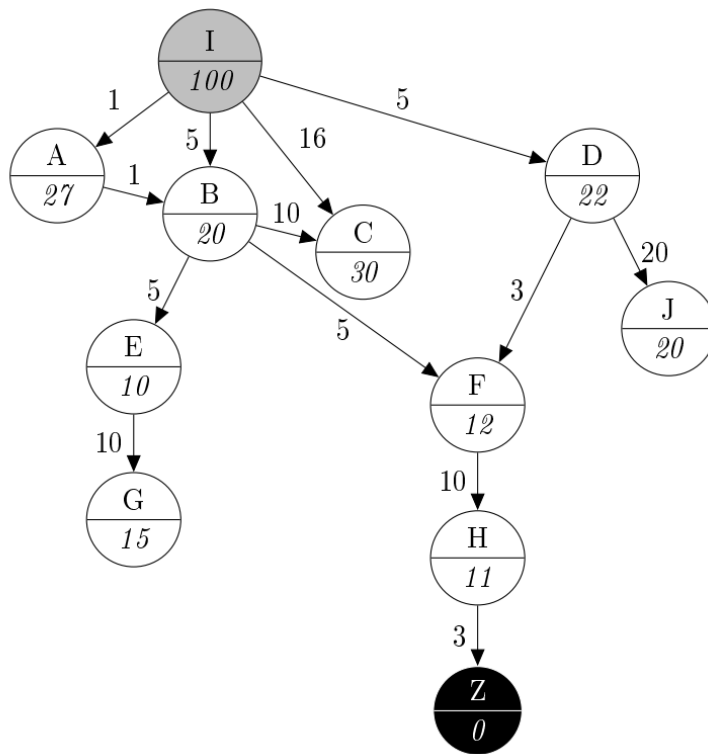
(M, V, 892)

~~(M, Z, 836)~~~~(S, B, 816)~~~~(B, Z, 912)~~

(Z, Bar, 860)

Cerrados~~(-, P, 580)~~~~(P, M, 789)~~~~(P, S, 808)~~~~(S, B, 816)~~~~(M, Z, 836)~~~~(Z, Bar, 860)~~

20.



Abiertos

~~(-, I, 100)~~
~~(I, A, 28)~~
~~(I, B, 25)~~
~~(I, C, 46)~~
~~(I, D, 27)~~
~~(B, C, 45)~~ → (B, C, 42)
~~(B, E, 20)~~ → (B, E, 17)
~~(B, F, 22)~~ → (B, F, 19)
 (E, G, 35)
~~(F, H, 31)~~ → (F, H, 29) → (F, H, 28)
~~(D, F, 20)~~
 (D, J, 45)

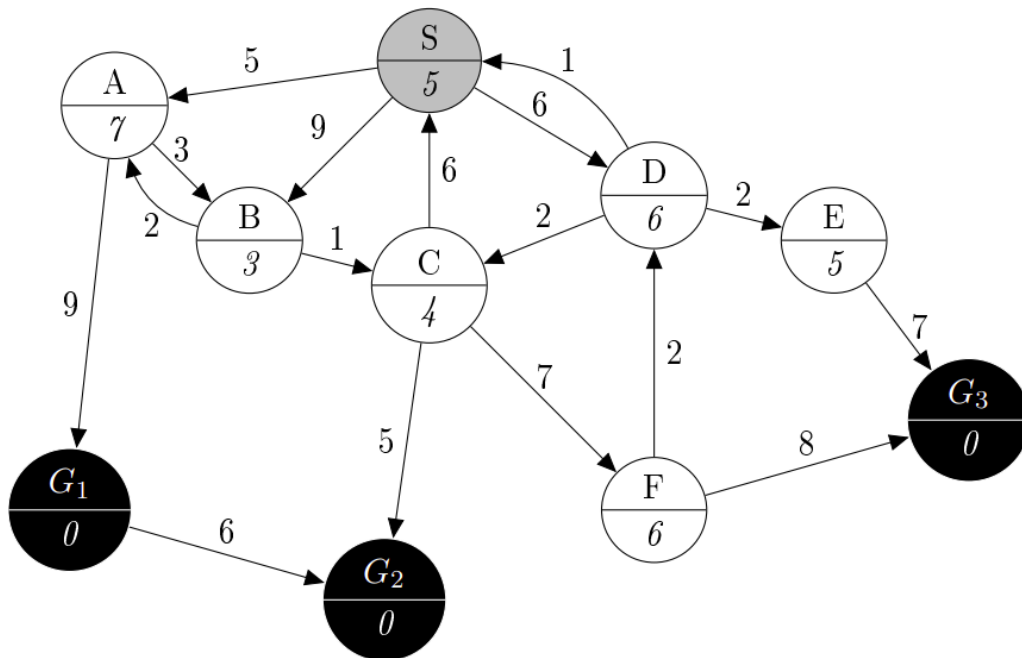
Cerrados

(-, I, 100)
~~(I, B, 25)~~
 (B, E, 20)
~~(B, F, 22)~~
 (I, D, 27)
~~(D, F, 20)~~
 (I, A, 28)
~~(A, B, 22)~~
~~(F, H, 29)~~
~~(B, F, 19)~~
~~(F, H, 28)~~

Camino: I → A → B → F → H → Z

Coste = 1 + 1 + 5 + 10 + 3 = 20

21.



a) Anchura

Abiertos
~~(-, S)~~

(S, A)

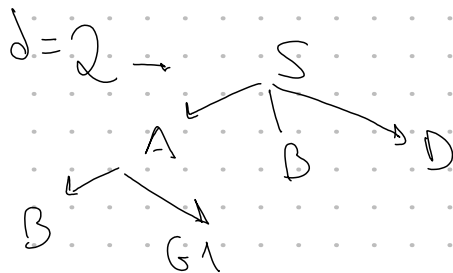
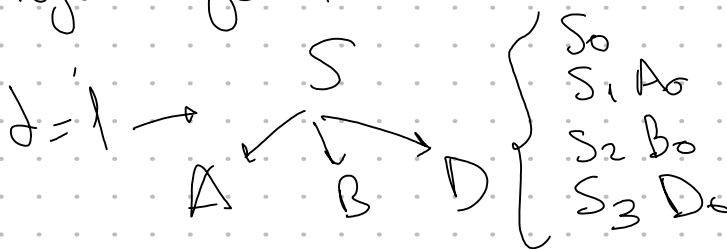
(S, B)

(S, D)

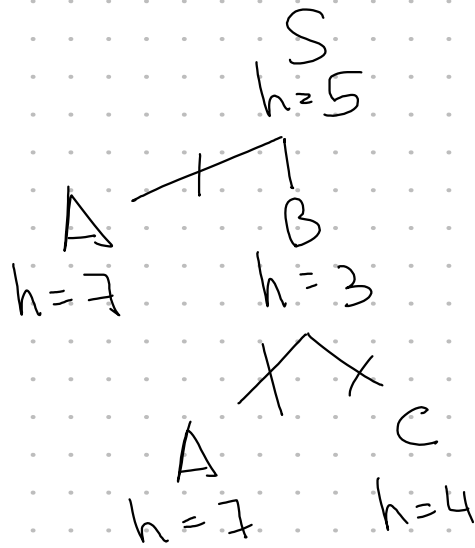
(A, G1)

Cerrados
 (-, S)

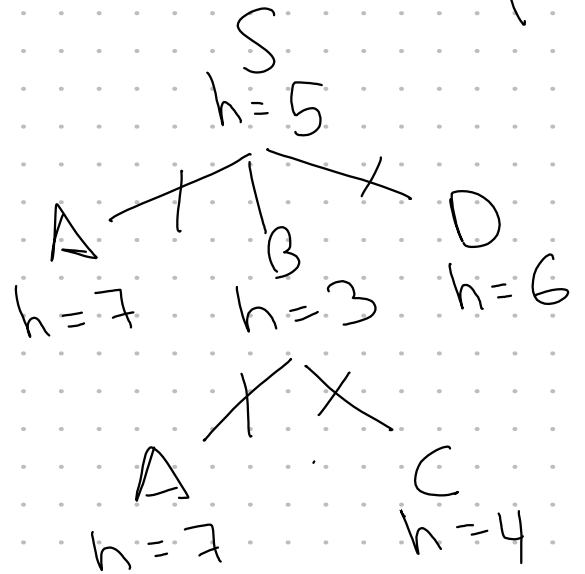
b) Profundizaje iterativo



c) Escalada simple



d) Escalada máxima pend.



e) A*

Abiertos

~~(-, S, 5)~~

~~(S, A, 12)~~

~~(S, B, 12)~~

~~(S, D, 12)~~

~~(A, B, 11)~~

(A, G1, 14)

~~(B, C, 13)~~

(D, C, 12)

~~(D, E, 13)~~

(C, G2, 13)

(E, G3, 15)

Cerrados

(-, S, 5)

(S, A, 12)

(A, B, 11)

(S, D, 12)

(B, C, 13)

(D, E, 13)

(C, G2, 13)